BIOCHEMICAL COMPOSITION OF CELL

Objectives

- Define the term biochemistry
- Describe cell and cell organelles
- Give functions of each organelles
- Give chemical composition of cell
- Justify their importance in life

What is Biochemistry?

- Bio means life
- Chemistry means chemical reaction occurs in living organisms

CELL AND CELL ORGANELLES :-

- All organisms are built from cell.
- All animals tissues including human are organized from collection of cells.
- Thus cell is fundamental unit of life.
- If cell dies then tissue also dies and can not function.

- Types of cells.
- The Prokaryotic cell.
- The Eukaryotic cell.
- Prokaryote cell:- a typical prokaryote cell include bacteria, smaller in size. do not have any membrane bound organelles. Genetic material not enclosed by nuclear membrane.

- Eukaryote cell: Include fungi, plants, animals including human, larger in size having considerable degree of internal structure have distinctive membrane with specific function.
- well defined nuclear membrane with genetic material in the form of chromatin.
- Reproduction involve mitosis and meiosis.

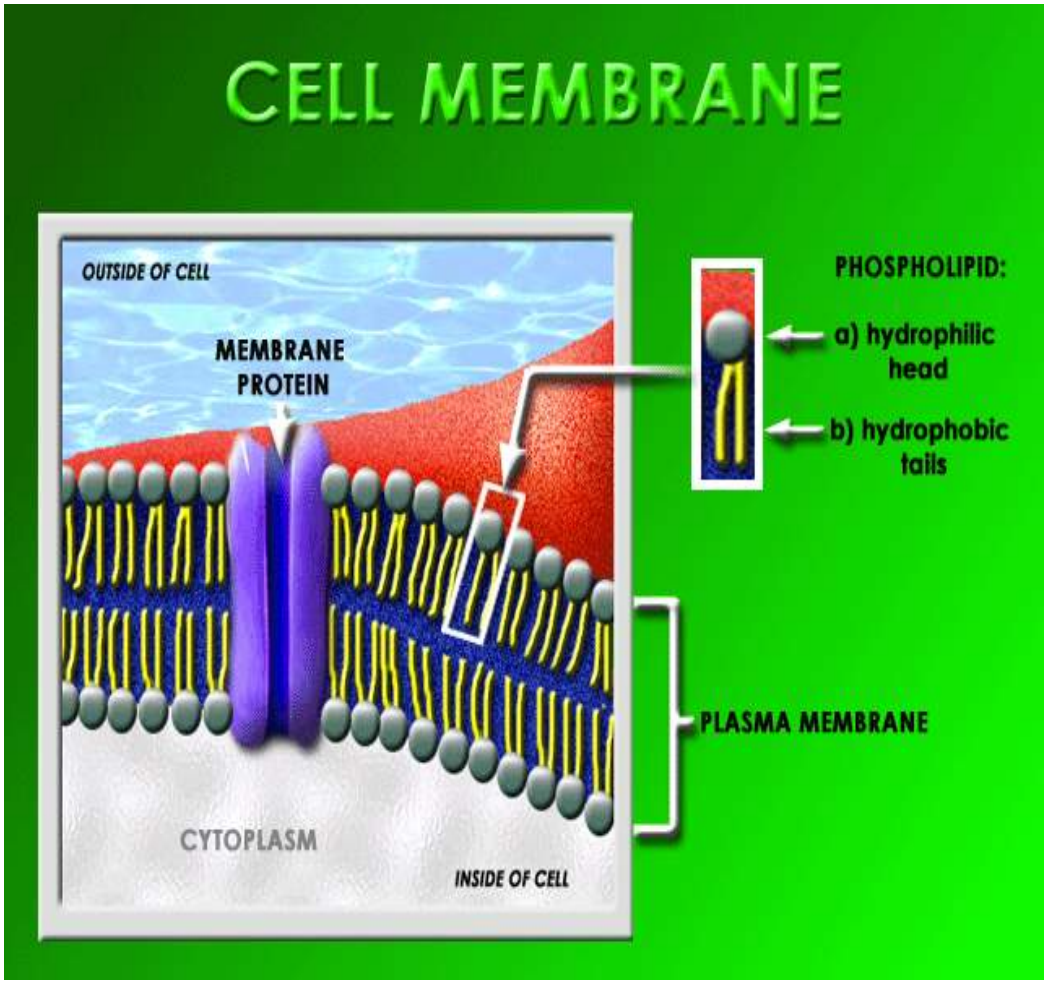
CELL ORGANELLES

- Eukaryote cell contain many membrane bound organelles with specific function such as:-
- 1) cell membrane:- membranes are composed of lipids, protein and carbohydrates.

- The relative content of these component varies widely from one membrane to another but typically it contain 40% of the dried weight is lipids about 60% protein and 10% of carbohydrate.

- Lipid molecule have a polar or ionic head hence hydrophilic and the other end is hydrophobic tail.

Cell Membrane



- Outer boundary of cell
- Protects the cell
- Controls what enters and leaves the cell

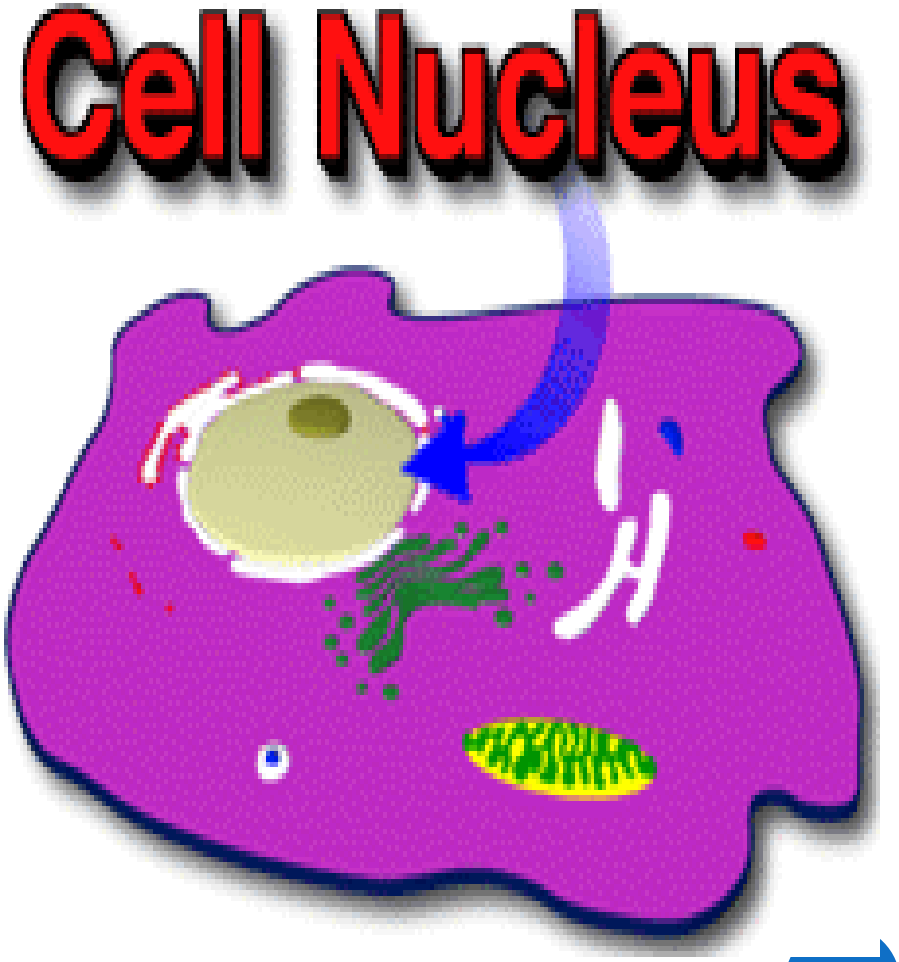
- NUCLEUS :- Nucleus contain more than 95% of cells DNA's and is the control center of the cell.
- Nuclear membrane separates nucleus from cytosol.
- Nuclear pores control the movement of protein molecules and nucleic acid across the nuclear membrane.
- Chromatins : DNA in nucleus is coiled dense mass called chromatin as it stain with certain dyes.

- Nucleolus a dense mass closely associated with inner nuclear membrane.
- Nucleoplasm: It contains various enzymes such as: DNA, Polymerase and RNA Polymerase for mRNA and tRNA synthesis.
- Function: DNA replication and RNA transcription of DNA occur in nucleus.

- Nucleolus is non membranous and contains RNA polymerase, RNAase, ATPase but not DNA Polymerase.
- Nucleolus is the site of synthesis of Ribosomal RNA (rRNA) Nucleolus is also the major site where ribosome's sub units are assembled.

The Nucleus

- Controls most activities in the cell
- Contains all genetic information in the form of DNA.



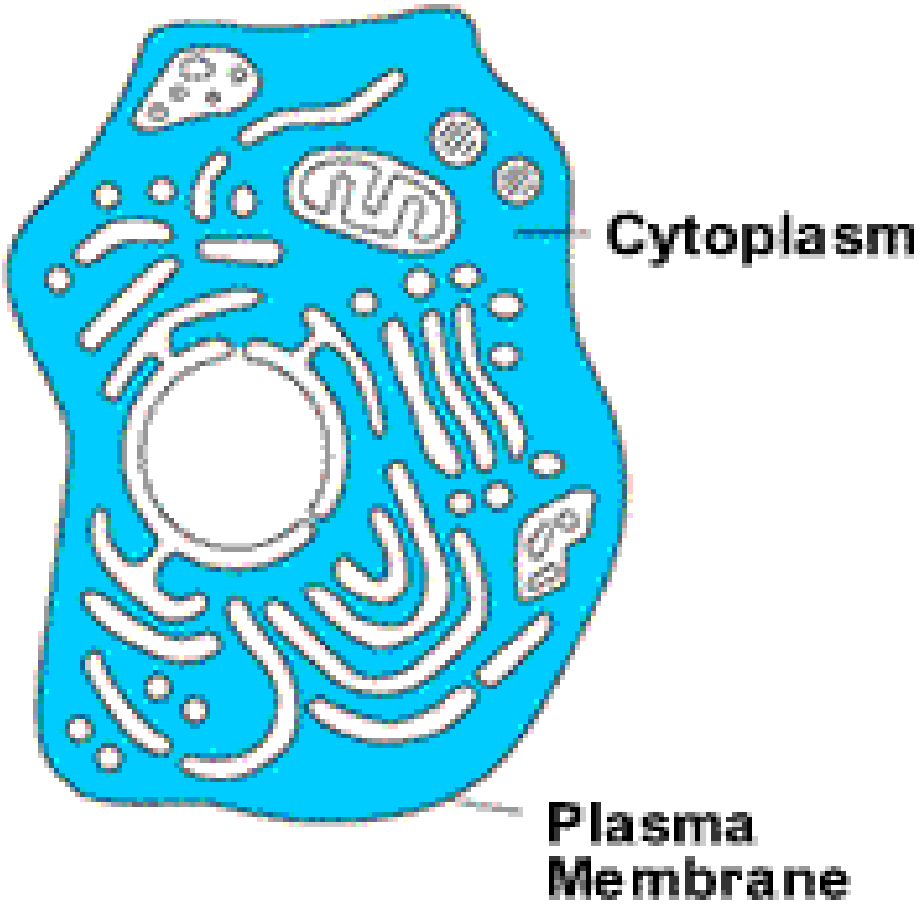
- CYTOPLASM:- This is simplest structure of the cell organelles many metabolic reaction take place in cytosol where substrates and co-factors interact with various enzymes.
- There is no specific structure for cytosol it has a high protein content major role of cytosol is to support synthesis of protein on the rough endoplasmic reticulum by supplying energy and co-factor.



- Cytosol also contain free ribosomes often in the polysome form they contain many different type of protein and ribosomal RNA they exist as to sub-unit and act as a site of protein synthesis.

Cytoplasm

- Clear fluid within cell that contains all organelles
- Moves materials throughout the cell

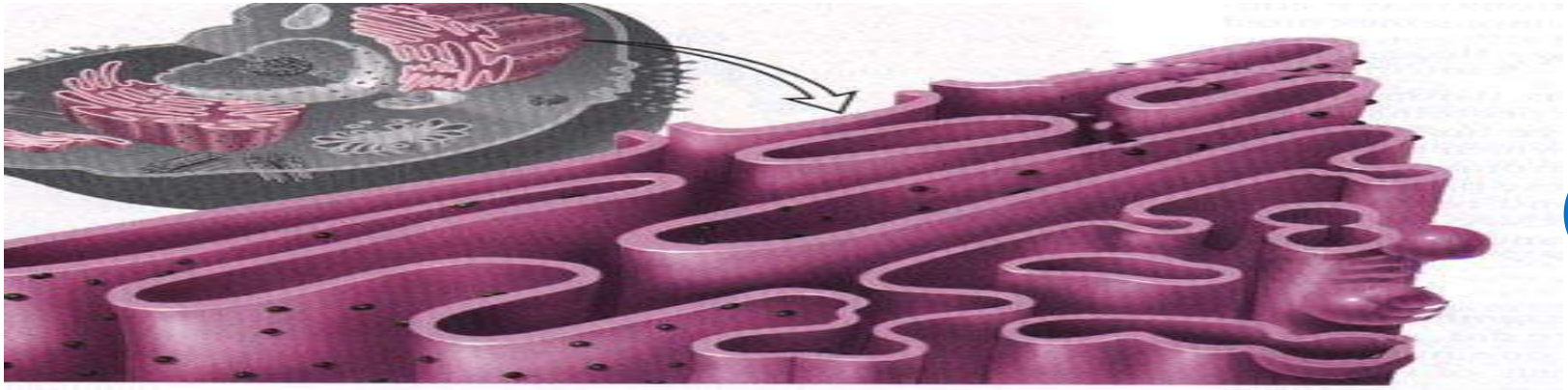


- ENDOPLASMIC RETICULUM: Cell are characterized by several membrane complexes that are interconnected by separated organelles they are involved in protein synthesis, transport, modification, storage and secretion.
- Wearing in shape, size and amount the endoplasmic reticulum extends from the cell membrane coats the nucleus surround the mitochondria and appears to connect directly to be Golgi apparatus the aqueous channel enclose are called cisternae.

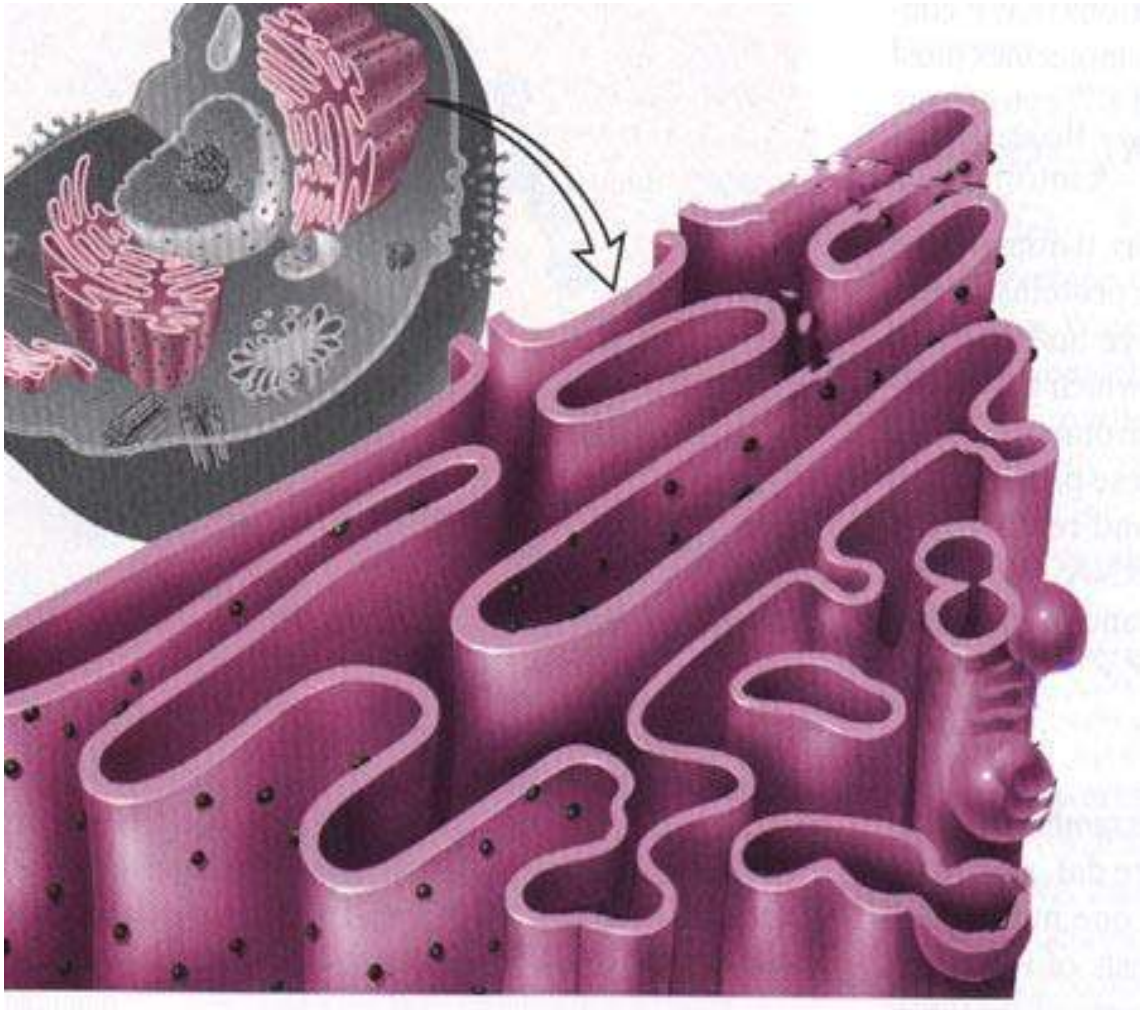
Types: they are two kinds,

1). Rough surface endoplasmic reticulum they are coated with ribosomes near the nucleus. It merges with the outer layer of the nuclear membrane.

2). Smooth surface endoplasmic reticulum:- they do not have attached ribosomes.



Endoplasmic Reticulum



- Series of folded membranes that form sacs or tubes
- Rough ER has ribosomes attached

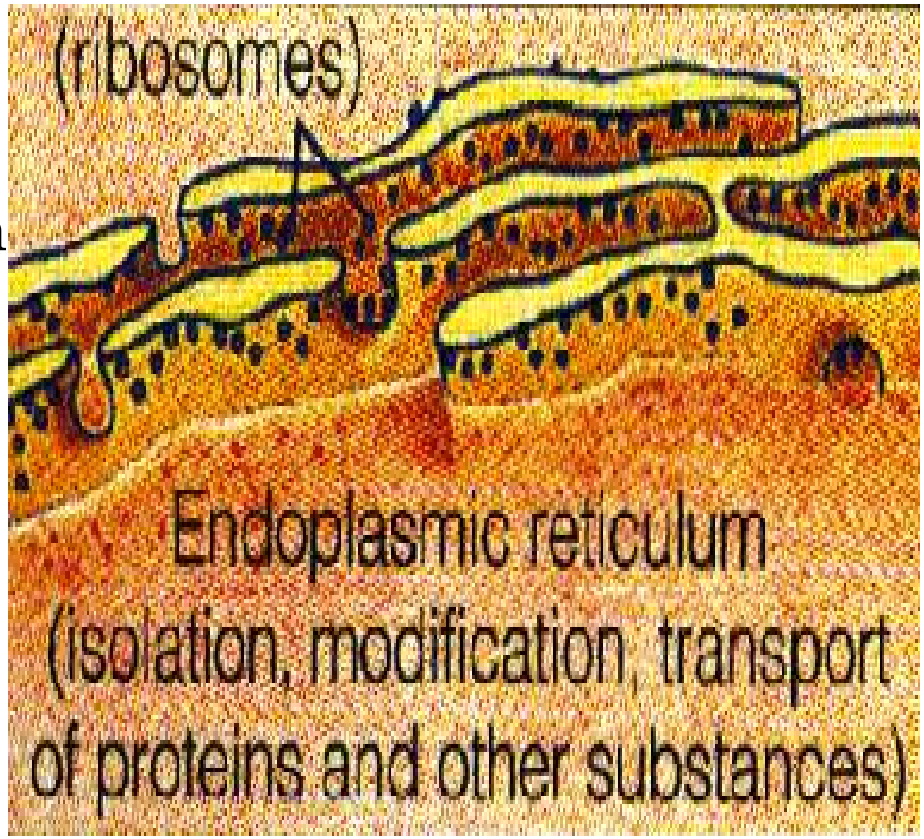
- Functions:
- Rough ER synthesizes membrane lipids and secretary proteins.
- These proteins are inserted through the ER membrane into the lumen of cisternae where they are modified and transported to the cell.



- Smooth ER involved in lipid synthesis and modification and transport of protein synthesize in the rough ER.
- Number of important enzymes are associated with ER of liver cell these include enzymes responsible for synthesis of triacylglycerol phospholipids and enzyme involved in detoxification of drugs.

Ribosomes

- Transports materials between the cytoplasm and nucleus
- Makes proteins in the cell
- May be free in cytoplasm or attached to ER
- Proteins are vital to life- all cells must produce them



GOLGI COMPLEX:

- Cell contain a unique mass of smooth surface compartment of cisternae and that makes up the Golgi bodies the golgi complex has proximal compartment, a medial compartment and a distal compartment it receive newly synthesize protein from ER.

- Function: On proximal site Golgi complexes receive newly synthesized protein from ER via transfer vesicles.
- Post Translational Modification take place in medial part where carbohydrate and lipid precursors are added to proteins to form glycoprotein and lipoprotein respectively.

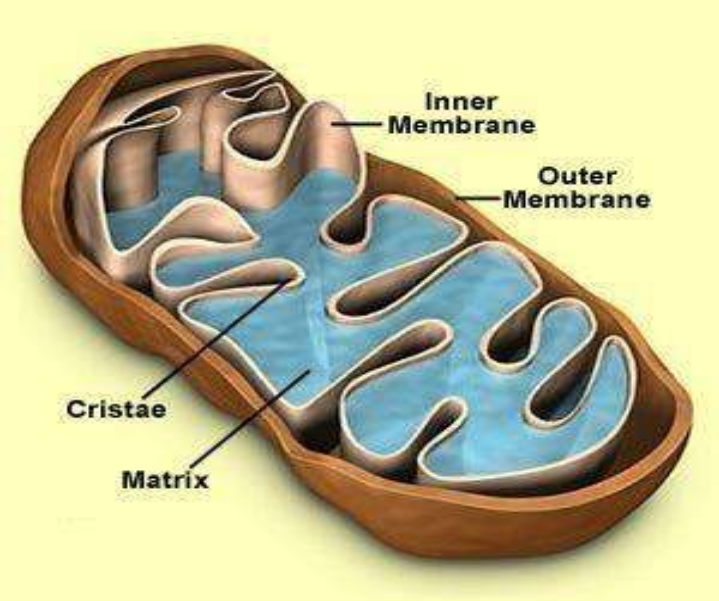
- On the distal site they release protein via modified membrane called secretary vesicles these secretary vesicles moves and fuse with plasma membrane where the content may be expelled by the process called exocytosis.

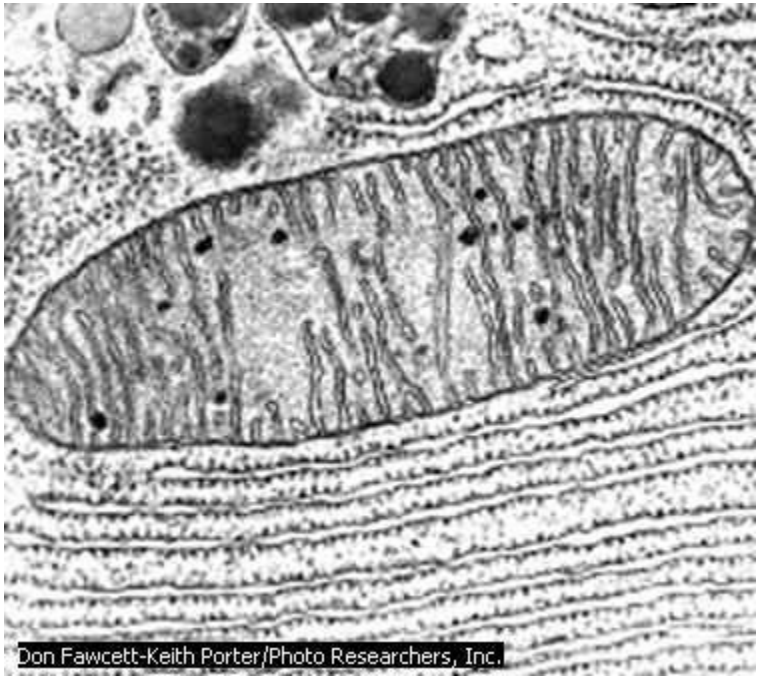
Golgi Bodies

- Saclike membranes used for storing/ packaging of chemicals
- Cells that make saliva or mucus have many Golgi bodies



- MITOCHONDRIA is a power house of the cell. Mitochondria is bounded by two concentric membranes that have markedly different properties and functions.
- Mitochondrial Membrane:
- 1: Outer mitochondrial membrane consist mostly of phospholipids and considerable amount of cholesterol and it also contain proteins, these protein form channels that permits substance less than molecular weight 10,000 to diffuse freely across the outer mitochondrial membrane.





- Energy producers
- Powerhouse of the cell
- Sausage-shaped
- Many mitochondria in muscle cells

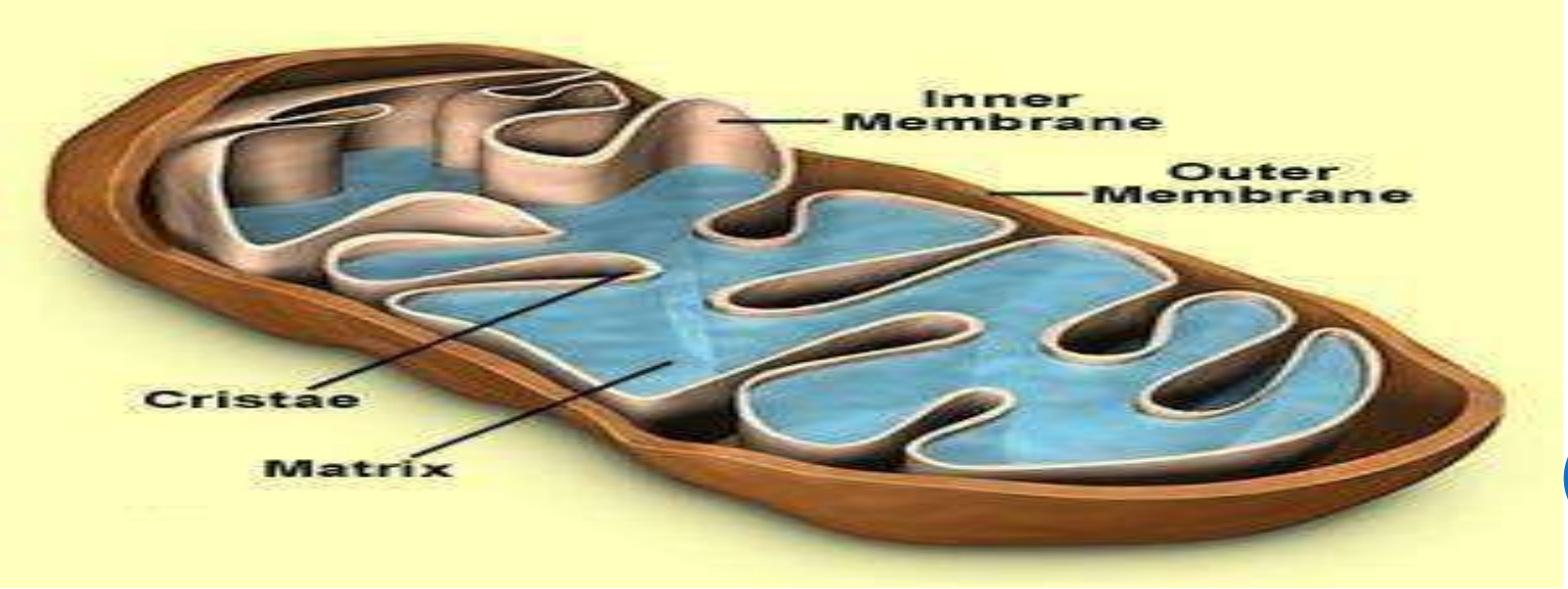
- Other protein in outer membranes carry out various reaction in fatty acid and phospholipid biosynthesis.

2: **Inner mitochondrial membrane** is very rich in protein.

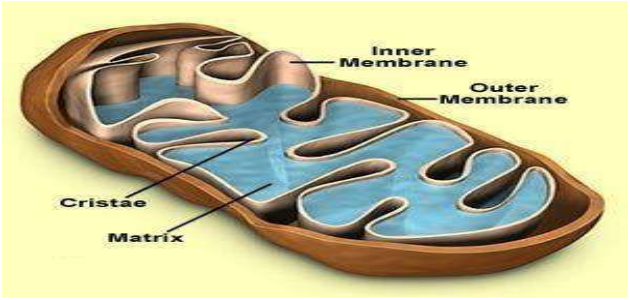
- It contain high proportion of the phospholipid cardiolpin.
- The inner mitochondrial membrane is impermeable to polar and ionic substance and they can be transported through protein.

INTER MEMBRANE SPACE:-

- The space between inner and outer membrane , since the outer membrane is freely permeable to small molecule the inter membrane space has the same composition as the cytosol.



MITOCHONDRIAL MATRIX:-



- The region enclosed by the inner membrane.
- Composition:- enzyme responsible for citric acid cycle and fatty acid oxidation are located in the mitochondrial matrix. It also contain several strand of circular DNA, ribosomes and the enzyme required for the biosynthesis of protein.

FUNCTION :-

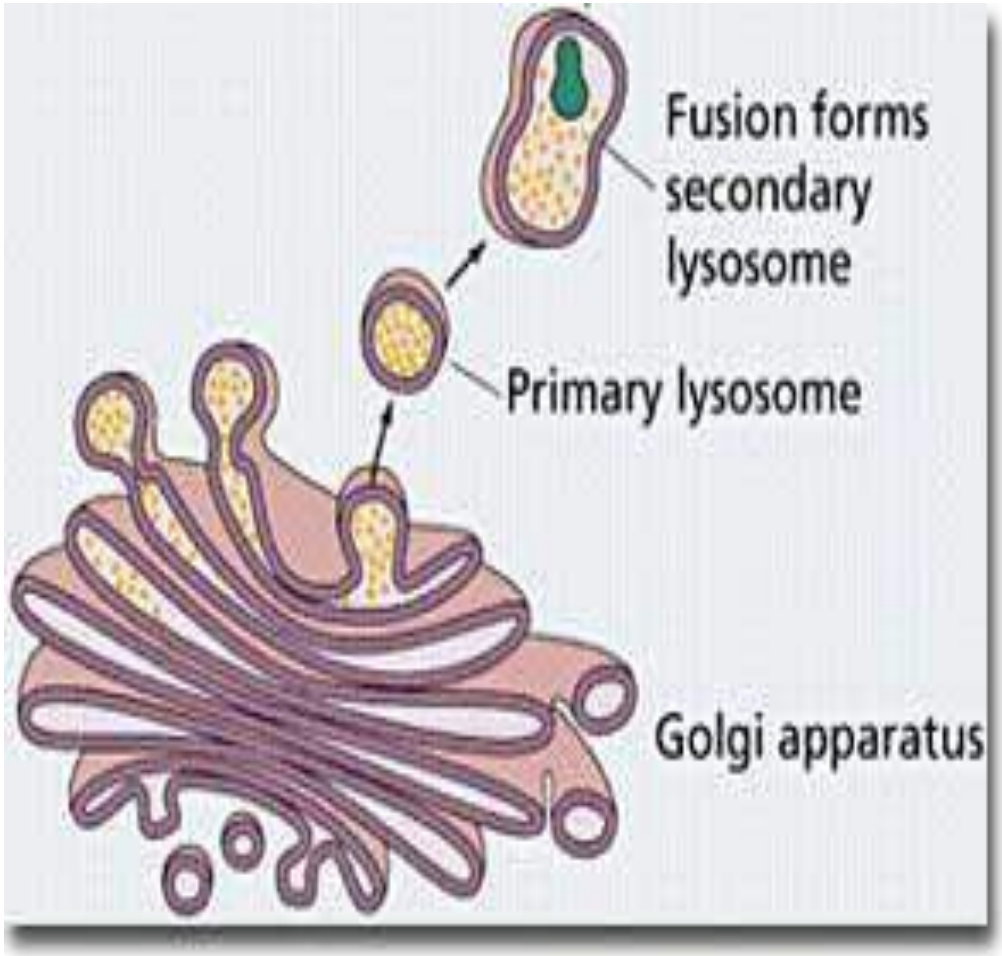
- Many enzymes associated with carbohydrates, fatty acid and nitrogen metabolism are located within the mitochondrion enzymes of electron transport chain and oxidative phosphorylation are also located here.



Location of some of the important enzymes in mitochondrion

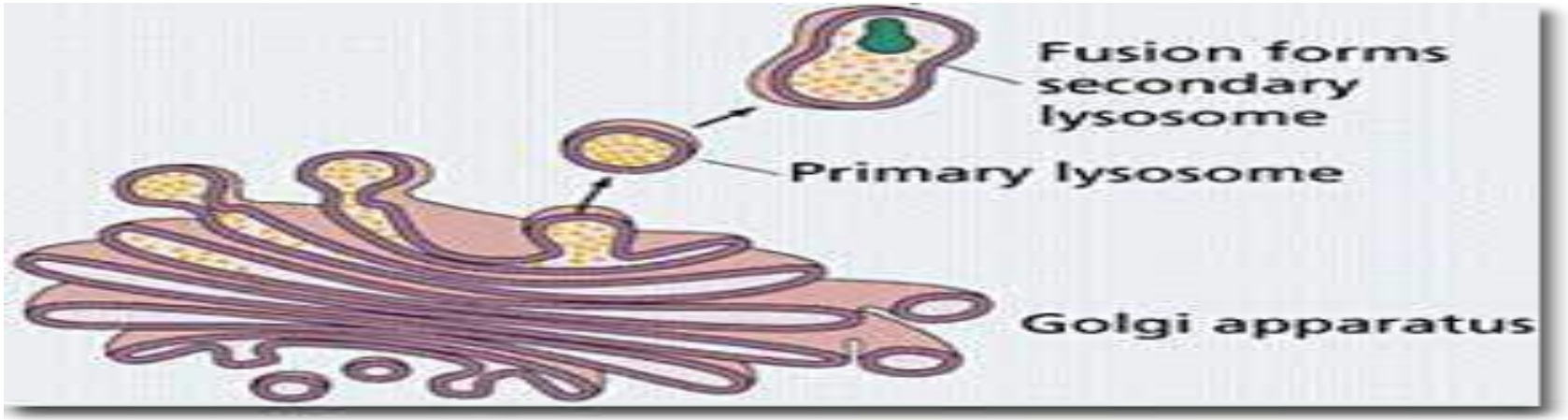
Outer membrane	Inter membrane space	Inner membrane	Matrix
Cytochrome b5 Cytochrome b reductase Fatty acid Co A synthase Phospholipase A Necleosida Diphosphokinase	Adenylate kinase Sulfite oxidase Nucleoside diphospho kinase	Cytochrome b, c, a, a_3 . NADH dehydrogenase Succinate dehydrogenase. Ubiquinone Electron transferring flavo proteins. ATP Synthase.	Pyruvate dehydrogenase . Citrate synthase Aconitase . Isocitrate dehydrogenase. Malate dehydrogenase. Carbamoylphosphate – synthase 1

Lysosomes

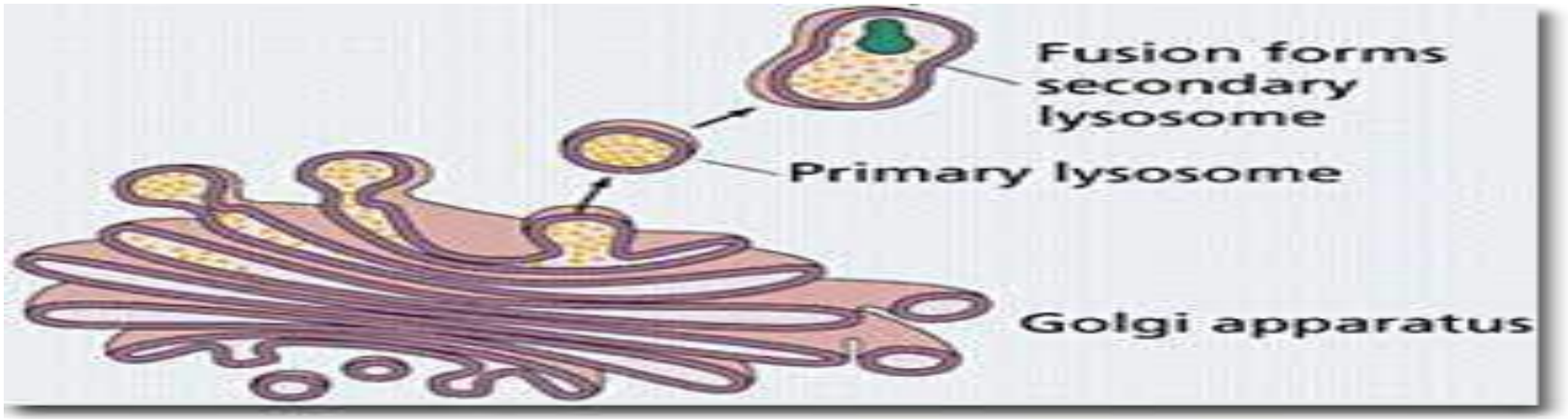


- Found in Golgi bodies
- Contain digestive enzymes to digest unwanted particles
- Help white blood cells to destroy bacteria

- Lysosomes contain packet of enzymes derived from greek word gree meaning lysis they are surrounded by lipoprotein membrane they are found in all animal cell except erythrocyte.
- pH inside the lysosome is lower than cytosol lysosomal enzymes have an optimal pH around 5, acid phosphatase is used as a marker enzyme for this organelle.



- Lysosomal enzymes essentially 30 to 40 are hydrolytic in nature.
- As long as the lysosomal is intact the enzyme can only act locally, but when the membrane is ruptured, the enzyme are released into the cytoplasm and hydrolyzed external substrates.



Lysosomal enzymes

1) proteolytic enzymes	Cathepsin(proteinase) Collegenase Elastase
2) Nucleic acid hydrolyzing enzymes	Ribonucleases Deoxyribonuclease
3) Lipid hydrolyzing enzymes	Lipase Phospholipases Fatty acyl esterase
4) Carbohydrate splitting enzyme	α- glucosidase β – galactosidase hyaluronidase


Thank you